# Leo Yao

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## Education

**Carnegie Mellon University** — August 2024 – May 2027
*B.S. in Computer Science and Statistics & Machine Learning* — Pittsburgh, PA

- **QPA:** 3.68/4.00
- **Relevant Coursework:** Computer Systems, Computer Security, Theoretical Computer Science, Functional Programming, Machine Learning, Imperative Computation (Data Structures & Algorithms)
- **Awards:** Dean's List, High Honors (Spring 2025, Fall 2025)

## Experience

**Teaching Assistant, Principles of Functional Programming (15-150)** — January 2026 - Present
*Carnegie Mellon University* — Pittsburgh, PA

- Led weekly labs for 20+ students and held regular office hours.
- Graded 300+ student submissions in SML every week, providing technical feedback on correctness, efficiency, and style of recursive, parallel, and higher-order algorithms.

**Research Assistant, CERN CMS Experiment** — September 2025 - Present
*Carnegie Mellon University* — Pittsburgh, PA

- Built custom PyTorch/HGQ2 GNN and its training pipeline for FPGA deployment at CERN CMS, achieving 97% reduction in loss compared to baseline model.
- Developing transformer-based particle clustering models and data pipeline for CERN HGCAL data.

**Research Assistant, SPICE Lab** — June 2025 – August 2025
*Carnegie Mellon University* — Pittsburgh, PA

- Engineered a Python data pipeline (Pandas, NumPy) to clean, normalize, and extract features from structured energy datasets.
- Optimized PyTorch neural networks using LightGBM feature selection and hyperparameter tuning, improving accuracy by 75% over initial baseline.

## Projects

**PharmaSentinel** | *Python, PostgreSQL, Supabase, OpenAI API, openFDA, MCP* — February 2026

- Engineered a multi-agent backend system in Python, orchestrating external REST APIs (OpenAI, openFDA) to automate real-time supply chain monitoring and substitute identification.
- Architected event-driven PostgreSQL databases utilizing Supabase Realtime to detect inventory state changes, automatically triggering agent pipelines and routing resolutions to downstream clients.

**Computer Systems** | *C, POSIX threads* — May 2025 – August 2025

- **Dynamic Memory Allocator**: Developed a dynamic memory allocator in C for Linux. Achieved 74.3% memory utilization and a throughput of 15885 kilo-operations per second, ranking at the top of the class.
- **Tiny Linux Shell**: Engineered a Linux shell featuring foreground/background job control, I/O redirection, and robust signal handling to manage concurrent process execution.
- **Shark File System**: Developed a thread-safe concurrent file system with standard features (e.g. file renaming, file descriptor position management), optimizing multithreaded performance and correctness with mutex synchronization.

## Skills

**Programming Languages:** Python, C, SQL, SML, x86-64 Assembly
**Backend & Systems:** PostgreSQL, Supabase, OpenAI API, MCP, Linux
**Data & ML:** PyTorch, Pandas, NumPy, scikit-learn
**Developer Tools:** Git, GDB, Valgrind